

Besher Alkurdi

AI Engineer · LLM systems: fine-tuning, deployment, and inference

Python · PyTorch · Hugging Face Transformers · vLLM · DeepSpeed · FSDP · GCP / Vertex AI · FastAPI · Docker · CI/CD

github.com/mrbesher linkedin.com/in/mrbesher huggingface.co/mrbesher Google Scholar YouTube Finland · Full work rights

EXPERIENCE

AI Engineer at University of Jyvaskyla — Nov 2025 - July 2026

- Migrated a multimodal AI pipeline for video and audio analysis from on-premise to Google Cloud Platform; worked on the async Python/ FastAPI backend with Cloud Run, Cloud SQL, GCS, Cloud Tasks, Cloud Scheduler, Secret Manager, and Vertex AI.
- Re-implemented the web frontend (Vue/TypeScript) with CI/CD via GitHub Actions and Firebase Hosting.
- Managed the NLP course for AI students, including project supervision and assessment.

AI Engineer at Huawei — Aug 2023 - Aug 2024

- Enhanced distributed training library with parameter-efficient training, context expansion, and stabilization techniques using ZeRO, FSDP, DeepSpeed, achieving 8x resource reduction.
- Optimized model architectures for open-source library compatibility and built scalable multi-backend inference platform.
- Developed multilingual synthetic data generation toolkit and fine-tuning pipelines for customer adaptations.
- Conducted requirements analysis and defined PoC roadmaps with cross-functional teams.
- Received New Employee of the Year, Future Star, and other awards for my contributions and training sessions I gave across Huawei Turkey.

Undergraduate Research Fellow at Yildiz Technical University — Jun 2022 - Mar 2023

- Curriculum learning research in publicly funded project using TensorFlow, PyTorch, and Horovod for difficulty-based training strategies.
- Trained CLIP-style multimodal models with contrastive learning and experimented with auxiliary classifiers for early exit strategies.

Research Intern at New Mind AI — Jul 2021 - Oct 2021

- Researched SoTA NLP techniques for a Document AI system assisting lawyers with Turkish legal documents.

EDUCATION

MSc. Artificial Intelligence at University of Jyvaskyla (GPA: 4.86/5) — May 2026

Thesis: The Kinematics of Cross-Lingual Transfer in Decoder Large Language Models

BSc. Computer Engineering at Yildiz Technical University (2nd in class, GPA: 3.84/4) — Jun 2023

Thesis: Large Language Models for Data Augmentation using Reinforcement Learning Techniques

PUBLICATIONS

ModernBERT-TR: A Modern Encoder Foundation Model for Turkish *Preprint* — Jul 2026

Scaling Up Paraphrase Generation Datasets with MT and Semantic Similarity Filtering — 2023

Semantic Similarity Based Filtering for Turkish Paraphrase Dataset Creation — Dec 2022

Linguistic-based Data Augmentation Approach for Offensive Language Detection — 2022

VOLUNTEER

Volunteer Researcher at COSMOS AI Research Group — Dec 2024 - Present

- Co-trained state-of-the-art Turkish text encoder; researching LLM/VLM improvement using unlabeled data and RL.

Tutor at Schoolhouse.world — Feb 2021 - May 2021

- Tutored students worldwide in Math, Computer Science, and related subjects.

Tutor at Aqsa Youth Institution — Nov 2020 - Aug 2021

- Taught computer, networking, and programming basics to middle and high school refugee students.

Translator at Ask Me — Jun 2019 - Aug 2019

- Assisted tourists visiting Istanbul as part of Fatih Municipality volunteer program.

CERTIFICATES

Large-Scale Training (Maven) — Oct 2025

Deep RL Course (HuggingFace) — May 2024

ML, Deep Learning, NLP & GANs Specializations (DeepLearning.AI) — 2022 - 2023

LANGUAGES

Arabic (Native) - English (Fluent) - Turkish (Fluent) - Finnish (Intermediate) - French (Beginner)